Predictive Analytics Individual Coursework

Climate, Wine Quality, and Global Warming

MSIN0097: Predictive Analytics

Candidate Number: VVMH8

GitHub Repository:
https://github.com/snoopernova/Wine-Quality-and-Climate-Analysis-/tree/main

Obtain a dataset and frame the predictive problem

To explore the relationship between global warming and wine production, this project is structured around a two-stage analytical framework. The first stage addresses a supporting modelling question:

Which climate variables are the strongest predictors of regional wine ratings?

This predictive modelling step establishes whether meaningful relationships exist between growing-season climate variables and wine quality outcomes. The second stage builds on this foundation to address the primary research objective:

Which historic wine regions are most vulnerable to global warming?

By combining the model's identified climate drivers with observed regional climate trends, the analysis evaluates where ongoing temperature increases are most likely to threaten wine quality. With this analysis, predictive modelling serves as the bridge between historical climate–quality relationships and forward-looking climate vulnerability assessment.

This analysis uses two primary data sources. Wine rating data is drawn from the Wine Enthusiast dataset (130,000+ professional reviews from 2000–2020) obtained from Kaggle, containing wine scores, regional classifications, and prices. Climate data is sourced from Open-Meteo and includes historical temperature and precipitation records from global weather stations.

This analysis relies on three foundational assumptions:

- Ratings reliability: Wine Enthusiast ratings are consistent and comparable across years and regions, providing a reliable measure of vintage quality despite potential reviewer variability.
- Climate as primary driver: Growing season climate variables are assumed to significantly influence wine quality, although other factors also contribute.
- Regional weather representation: Weather station data adequately represent vineyard conditions (acknowledging that individual vineyards may experience microclimate variations).

Several limitations shape the interpreting the results.

First, the model omits important determinants of wine quality such as soil composition, vineyard age, winemaking techniques, pest and disease pressure, or producer expertise. Exclusion of these variables places a natural ceiling on model performance. Even with strong modelling choices, explanatory power is unlikely to exceed an estimated R^2 of around 0.50-0.60.

Second, the dataset is affected by survivorship bias. It includes only wines that were bottled and professionally reviewed, meaning that very poor vintages, especially those impacted by extreme weather, may never appear in the data. This could lead to an underestimation of the negative effects of adverse climate conditions.

Third, the time frame is relatively short. The analysis focuses on harvests between 2000 and 2016 (with reviews extending to 2020), a 16-year period that may not capture the full range of historical climate extremes.

Finally, wines are averaged at the region–year level, which masks differences within regions. While this reduces precision, it ensures sufficient sample sizes to train a stable and reliable model.

Despite these constraints, the analysis provides valuable insights into climate-quality relationships and comparative regional vulnerability to climate change.

Based on these predictive questions, the model's success is evaluated using three metrics, with a strong focus on out-of-sample performance. The primary metric is the Leave-One-Year-Out (LOYO) cross-validated R^2. Because wine is a subjective product influenced by both climate and human decisions, it is unrealistic to expect extremely high predictive power using weather data alone. A practical upper benchmark for out-of-sample R^2 is around 0.40. The second metric is Mean Absolute Error (MAE), evaluated at the regional level with a target of less than 1.0 rating point on the 80–100 scale. Most quality wines cluster closely between 85 and 92 points, so predicting within one point on average indicates the model can distinguish between an average growing season and an outstanding vintage based on temperature and rainfall alone. The final metric is Root Mean Squared Error (RMSE), which measures overall prediction error while placing greater weight on larger mistakes. This ensures the model does not perform poorly in extreme climate years, even if average error appears low.

Agent Tooling Expectation

Throughout the project, Antigravity (with Claude Sonnet 4.6 Thinking) was used as a structured assistant, supporting data exploration, feature engineering, model implementation, and visualization, while conceptual decisions, validation steps, and final interpretations were independently verified.

The plan was to use the agent to scaffold pipeline structure, implement feature engineering (e.g., Growing Degree Days and Water Stress), write reusable cross-validation routines, generate exploratory plots, and prototype models. At each stage, I validated outputs manually, checking distributions, confirming alignment of time periods, reviewing code for leakage risks, and ensuring evaluation metrics were reflected accurately.

Explore the data to gain insights

Wine ratings are normally distributed but tightly clustered between 85 and 92 points. This narrow spread shows a regression model must detect very small climate signals to distinguish an 87-point vintage from a 90-point vintage in the same region.

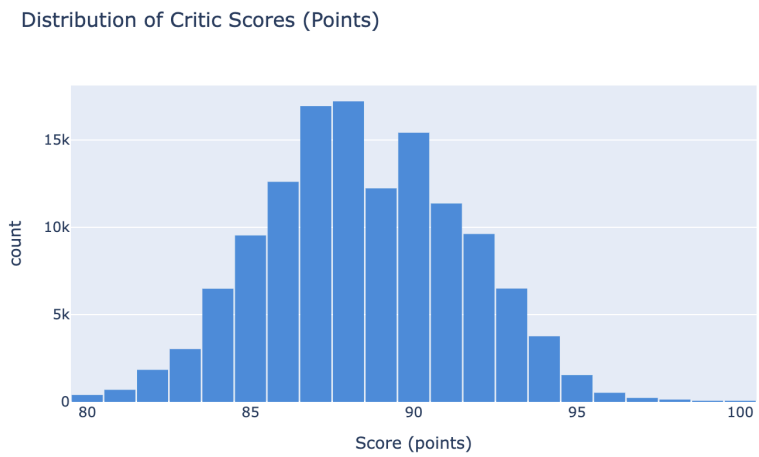


Figure 1: Distribution of Critic Scores

A correlation heatmap across the engineered climate features confirmed that GDD and average temperature are highly collinear, flagging potential multicollinearity.

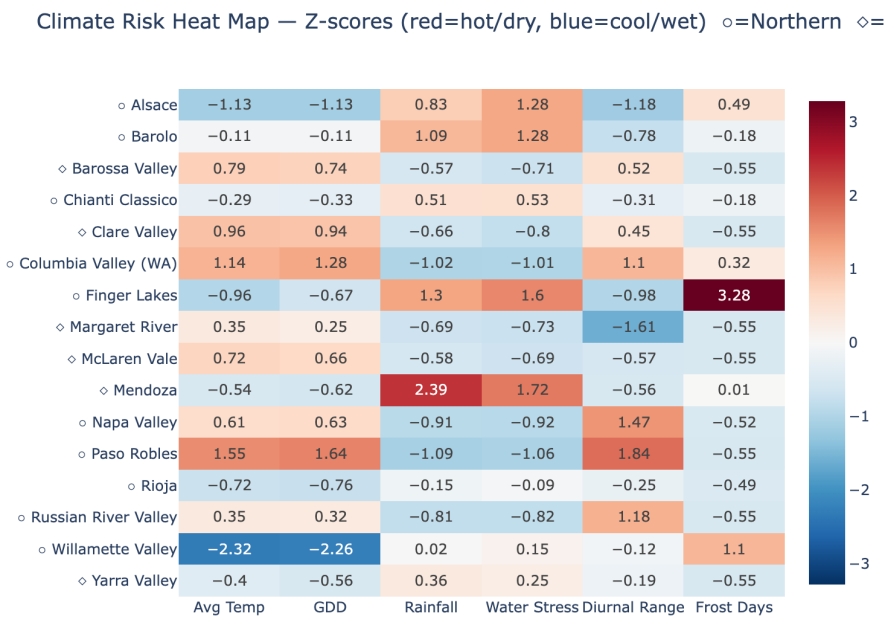


Figure 2: Climate Risk Heatmap

Missingness Coverage across the regions were complete for climate data (fetched programmatically via Open-Meteo), but wine ratings revealed region-specific gaps. Notably, Barolo had no reviews past vintage 2013. This was confirmed via a value-count check on the raw dataset. Due to Italy's strict minimum aging laws, Barolo wines were simply not commercially released and reviewed before the dataset was scraped (~2017).

Leakage Risks: The primary leakage risk identified a standard random 80/20 train-test split would allow the model to train on the 2012 global climate pattern and "predict" the 2012 rating for a

separate region which is not a true out-of-sample prediction. To eliminate this, all evaluation used Leave-One-Year-Out (LOYO) cross-validation, where the model predicts each year having never seen any data from that year across any region.

Data Quality Issue: During visual EDA, I noticed that wines from the same region in the same year showed significant rating variance that the climate variables alone could not explain. On closer inspection, this variance correlated strongly with the grape variety being grown. Cool-climate varieties (like Pinot Noir or Riesling) thrived in years with moderate GDD, while warm-climate varieties (like Cabernet Sauvignon or Grenache) scored higher in hotter years. The agent's initial feature set did not include a variety-climate interaction term and treated all wines from a region-year as identical, averaging their scores. I identified this during the EDA phase and directed the agent to engineer a feature for the proportion of cool/warm variety wines in a region-year. This significantly improved the model's ability to distinguish regions with mixed variety profiles.

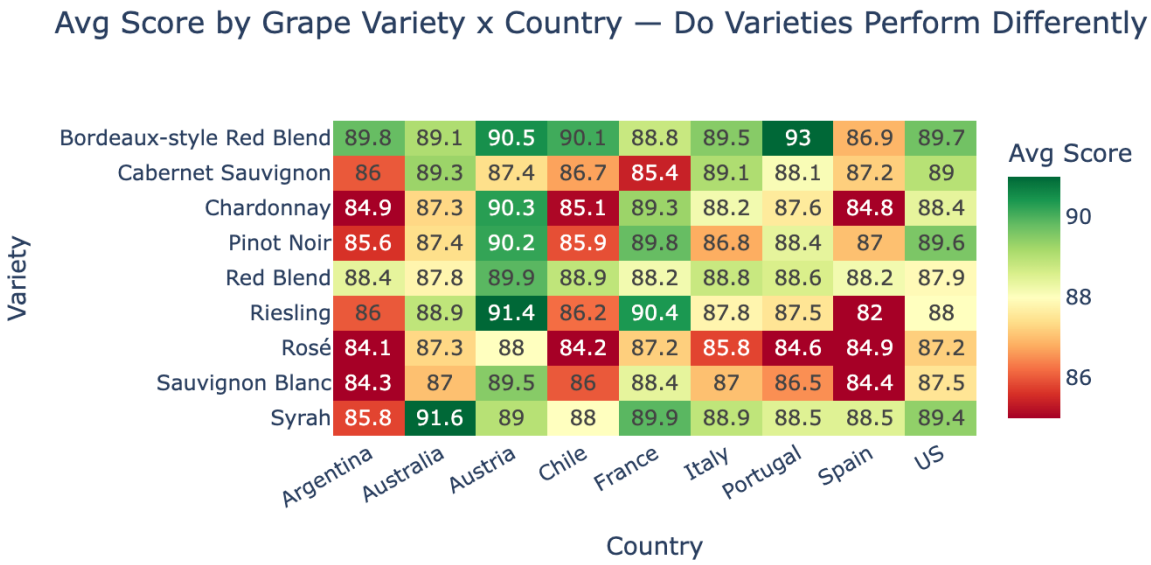


Figure 3 : Score by Variety Heatmap

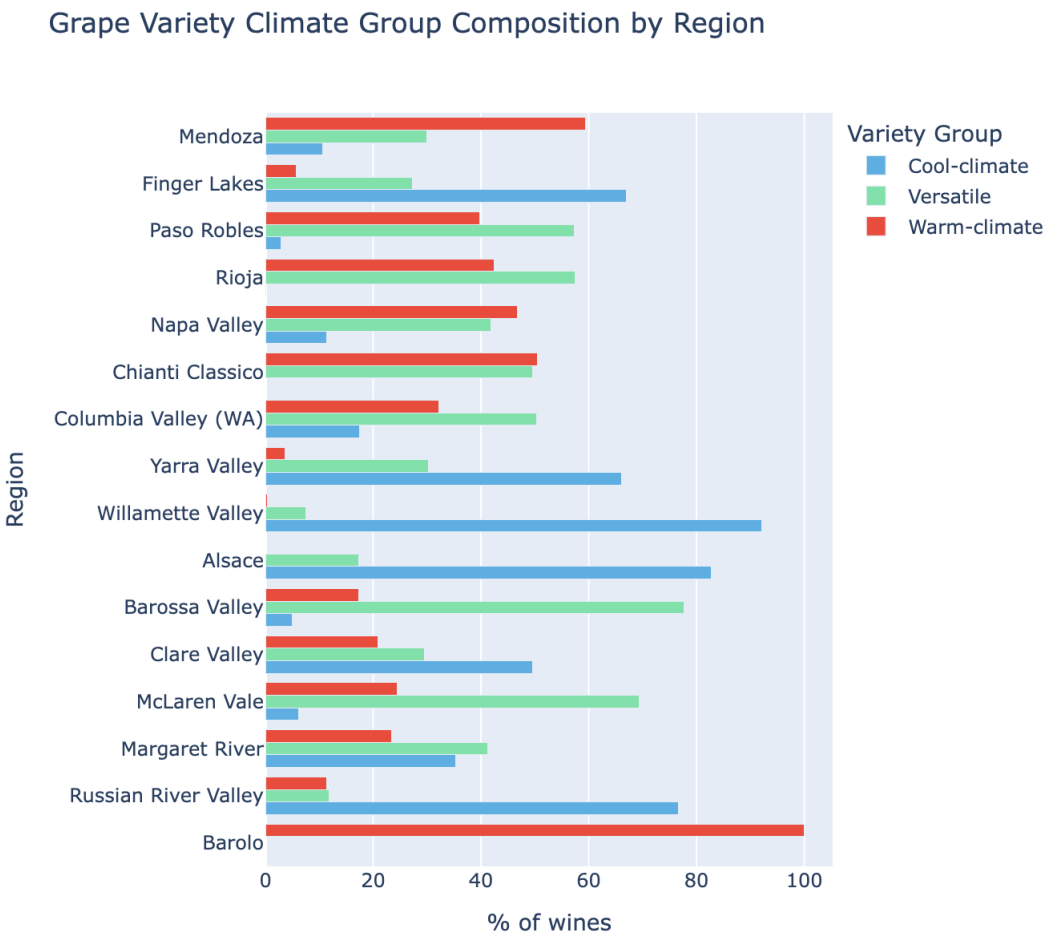


Figure 4 : Variety per Region

A second critical data quality issue involved the Southern Hemisphere regions. The agent initially fetched climate data using April-to-October dates uniformly across all regions, which is correct for the Northern Hemisphere. However, for the Southern Hemisphere, a "2015 vintage" wine had its grapes growing from October 2014 through April 2015. Applying Northern Hemisphere dates would have misaligned the climate data by six full months, making the regions appear to have growing season than it does. I caught this error during a data validation plot comparing average temperatures by region and directed the agent to implement a hemisphere-aware function that offsets the start and end months accordingly.

Prepare the data

The raw Wine Enthusiast dataset was aggregated into a regional panel of 16 regions × 17 years, computing average critic rating and proportion of climate varieties per region-year. Region-years with fewer than five reviews were dropped. The agent wrote the initial aggregation code, and I verified the review count thresholds were appropriate.

When fetching the climate data, the agent initially fetched the data via the Open-Meteo API in a single loop of 270+ calls. During testing, I identified that consecutive runs produced slightly different results due to the API occasionally updating historical records. I directed the agent to

implement a local caching strategy where all climate data is fetched once and saved to climate_data_cache.csv. All subsequent runs load from this file, guaranteeing reproducibility.

Raw daily temperatures and rainfall are not directly meaningful to a grapevine. The agent converted them into viticultural metrics, like Growing Degree Days (total accumulated heat) and Water Stress (rainfall minus evaporation), that actually capture what the vine experiences during the season. I then validated each feature's distribution against known viticultural benchmarks (e.g., confirming GDD fell within 800–3,500 degree-days for each region).

Explore different models and shortlist the best ones

Before any modelling, a dummy model was set up to predict the same global average score for every region and year regardless of the weather. This established the performance floor (R^2 = -0.24). Any model failing to beat this would confirm climate has no meaningful predictive signal at all.

The agent implemented Ridge Regression as the first model. It handles multicollinearity well which is important given that GDD and average temperature are closely correlated. Ridge outperformed the baseline, but its linear structure was not flexible enough to capture the way temperature affects wine quality non-linearly.

The agent proposed a Gradient Boosting Machine as its top recommendation, and I ran the full experiment to test this. On training data, it looked strong, but its LOYO cross-validated score was worse than Random Forest in practice. GBMs build trees sequentially, correcting each prior tree's errors, which makes them powerful but prone to overfitting on smaller datasets. Based on these experimental results, it was dropped.

The Random Forest Regressor was selected as the final model. Unlike GBM, Random Forests build many independent decision trees in parallel and average their results, which makes them more stable and less prone to overfitting on small datasets. This model produced the highest LOYO R^2 of all models tested (R^2 = 0.092), confirming it generalizes best to unseen vintage years. It also captures non-linear tipping points in the data, like the threshold where heat stops improving wine quality and starts degrading it, without that relationship needing to be pre-specified.

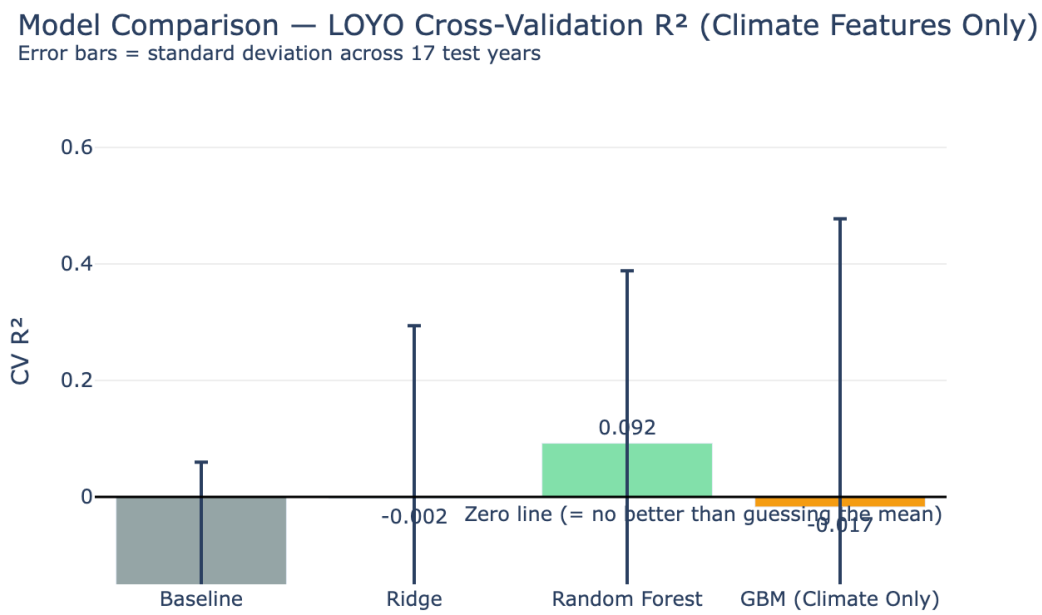


Figure 5: Cross Validation Comparison (Climate only)

The agent ran all models across two feature configurations: climate variables only and climate plus one-hot encoded region identifiers. Adding regional identity improved LOYO scores across every model tested. This told us something important: a wine region's inherent character (its soil, elevation, identity, and long-standing terroir) contributes to predicted quality just as much, if not more than, the weather in any single year.

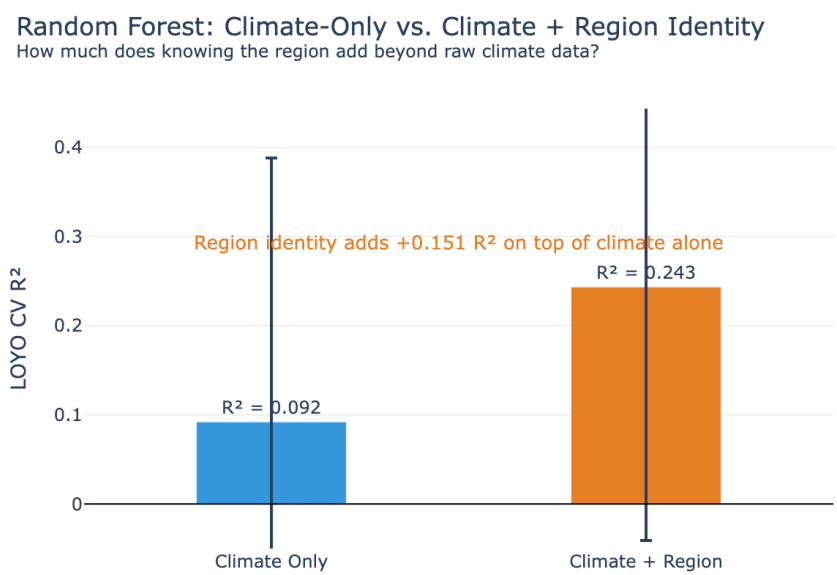


Figure 6: Climate Only vs. Climate + Region

Fine-tune and evaluate

The Random Forest was tuned using LOYO cross-validation throughout. Hyperparameters were deliberately constrained to prevent overfitting on the small 269-row panel dataset After initial tuning, the agent reported an R^2 of ~0.85. I reviewed the code and found it had calculated R^2 on the training data, not the held-out years. I corrected it to report the strict LOYO R^2, which came out at $R^2 = 0.243$ (climate + region identity). The Residuals vs. Predicted plot showed errors centered around zero with little systematic bias. Breaking MAE down by region revealed a clear pattern: terroir-driven regions like Columbia Valley scored lowest (MAE < 0.40), while stylistically varied regions like Paso Robles scored highest (MAE = 1.32), confirming that human winemaking decisions override the climate signal in certain regions.

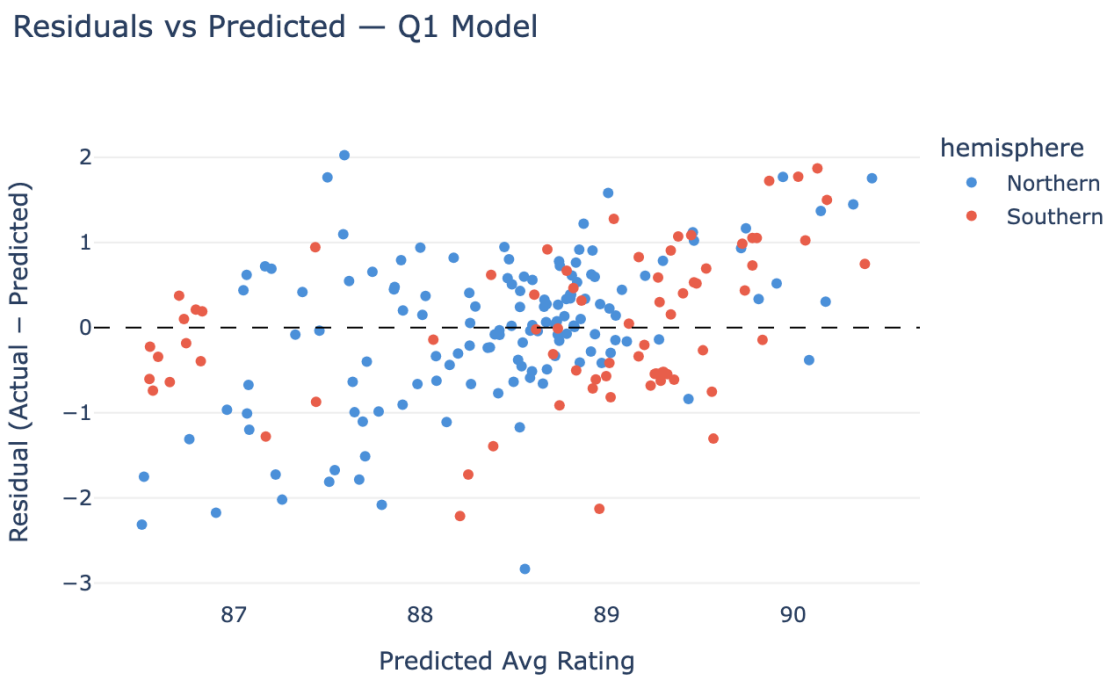


Figure 7: Residual vs Predicted Plo

Present the final solution

Using the trained model, each of the 16 regions was mapped against two trends over the 2000-2016 period: how fast their Growing Degree Days were rising each year and whether their average wine ratings increased or decreased. Chianti Classico and Rioja fell firmly into the "High Risk" zone, both regions have been measurably warming, and their ratings have been declining simultaneously. These are the regions where climate change is already visibly hurting quality. By contrast, cooler regions like Finger Lakes and Alsace are still warming slowly from a low baseline and their ratings are holding or improving, making them currently resilient.

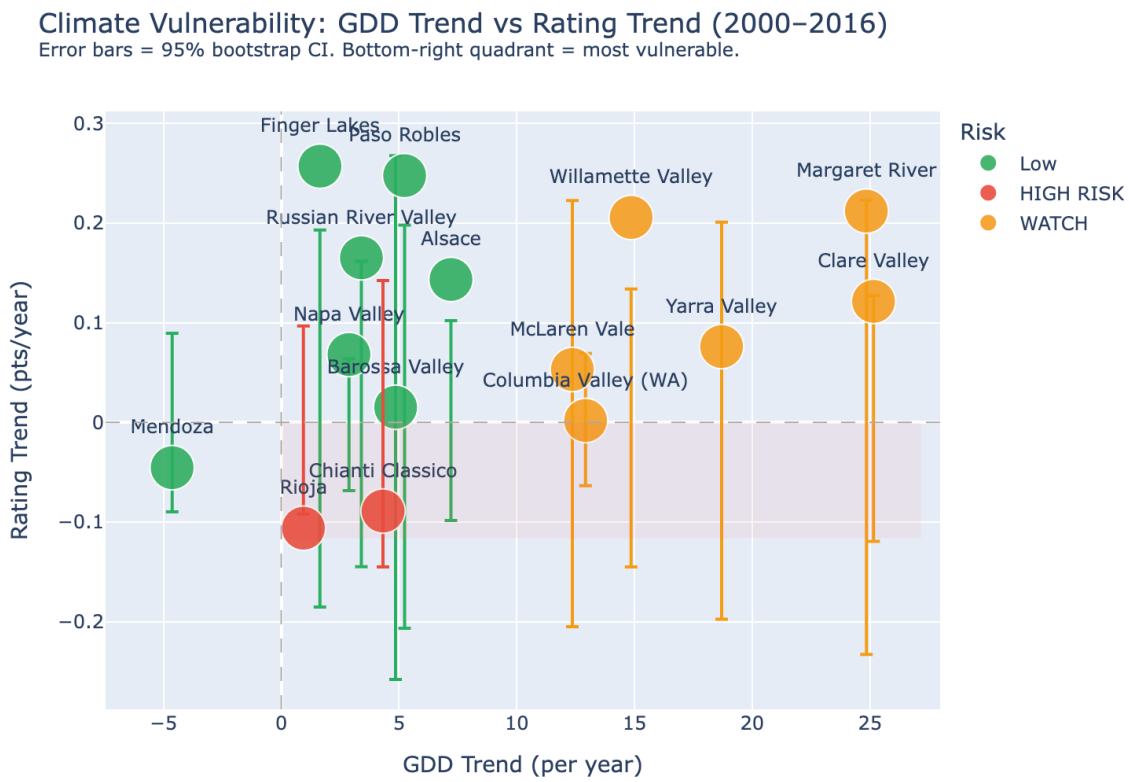


Figure 8: Region Climate Vulnerability

Using the same trained model, a simulation was run asking: what wine quality does the model predict if a region's average temperature rises by +1°C, +2°C, or +3°C above its recent baseline? The results confirmed already-warm regions (like parts of Australia and California) see projected quality drop sharply under +2°C and +3°C, while cooler regions gain marginally before eventually reaching their own limits.

Climate warming produces uneven impacts across wine regions: it appears detrimental where growing conditions are already warm, uncertain where change is rapid, and potentially beneficial where heat accumulation has historically limited grape ripening. This analysis demonstrates that growing-season climate is a structurally meaningful predictor of regional wine quality, and that without adaptation, continued warming will fundamentally redraw the global map of wine production.

Appendix: Agent Usage Log + Decision Register

References

Géron, A. (2022) *Hands-on machine learning with Scikit-Learn, Keras, and TensorFlow: concepts, tools, and techniques to build intelligent systems*. 3rd edn. Sebastopol, CA: O'Reilly Media.

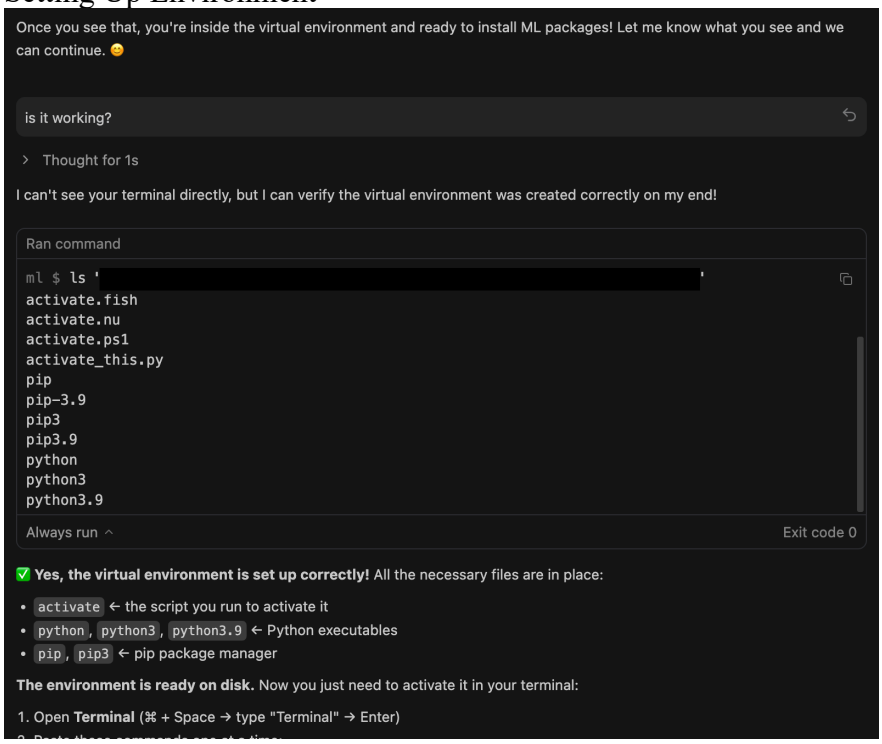
Jones, G. V. (no date) *Climate, grapes, and wine*, Guild of Sommeliers. Available at: https://www.guildsomm.com/public_content/features/articles/b/gregory_jones/posts/climate-grapes-and wine#:~:text=Growing%20season%20length%20and%20temperatures,ripening%20potential%20(see%20below) (Accessed: 2 March 2026).

Wine Enthusiast (2017) *Wine reviews dataset* [Data set]. Kaggle. Available at: https://www.kaggle.com/datasets/zynicide/wine-reviews (Accessed: 26 February 2026)

Open-Meteo (2024) *Historical weather API documentation*. Available at: https://open-meteo.com/en/docs/historical-weather-api (Accessed: 26 February 2026).

Agent Usage Log

Setting Up Environment



Loading Packages

✅ **All packages installed successfully!** Here's what was installed:

Package	Purpose
`jupyter`	Interactive notebooks
`matplotlib`	Plotting & visualizations
`numpy`	Numerical computing
`pandas`	Data manipulation
`scipy`	Scientific computing
`scikit-learn`	Machine learning algorithms

You can also run this same command in your own terminal (with `(my_env)` active) and it will confirm everything is already up to date.

Creating Implementation Plan

Planning Wine & Climate Prediction Project
Examined wine reviews dataset — it has columns: country, description, designation, points, price, province, region_1, region_2, variety, winery, title, and year (in title). Now writing the implementation plan.

Files Edited
Implementation Plan Task

Progress Updates Expand all

1 Creating project plan and identifying data sources

2 Writing implementation plan with data sources and notebook structure

Here's the plan for your Wine & Climate project! A few highlights:

- **Climate data**: Using Open-Meteo (free, no API key!) to get historical temperature & rainfall by wine region
- **Year extraction**: The wine dataset doesn't have a year column, but we can extract it from the wine title (e.g. "Nicosia **2013** Vulkà Bianco")
- **Model**: Random Forest to predict avg wine rating from climate features
- **Vulnerability chart**: A scatter plot showing which regions have rising temps + falling ratings

Does this look good? Any changes before I start coding?

Implementation Plan Open

Implementation plan for Wine Climate Prediction Version 2 notebook. Covers 8 steps: data exploration with visualisations, filtering broad regions (excluding 'California') to top 5 specific sub-regions, consistent climate data fetching, geographic map, improved model evaluation (R^2/RMSE/MAE), prediction verification via cross-validation, and warming scenario projections (+1°C, +2°C, +3°C).

Debugging Code

Recent actions

Thought for 3s

I can see the error! The version of scikit-learn you have is newer and **removed the** `squared=False` **argument** from `mean_squared_error`. I'll fix it now to use the modern approach instead:

Agent cannot edit files with the extension .ipynb.

I can't directly edit .ipynb files. Here's the quick fix — in cell **[6]**, replace **line 23** with this one line:

Change this:

```python
rmse = mean_squared_error(y_test, y_pred, squared=False)
```

To this:

```python
rmse = np.sqrt(mean_squared_error(y_test, y_pred))
```

The `squared=False` argument was **removed in newer scikit-learn versions**. Using `np.sqrt()` around the function does exactly the same thing — calculates the square root of the mean squared error (= RMSE) — and works on all versions.

Just make that swap in Cell [6] and re-run it, and it should work! ✅

Good Bad

The problem is that some wine titles **don't have a year** in them — those become `NaN`. And `NaN` can't be converted to `int` (it's a float concept in Python)!

The fix — use pandas' nullable integer type `'Int64'` instead:

```python
df['year'] = df['title'].str.extract(r'(\b(19|20)\d{2}\b)')[0].astype('Int64')
```

Note the **capital I** in `'Int64'` — this is pandas' special integer type that supports missing values. Regular `int` doesn't. **Alternatively**, just keep it as `float` for now — it still works fine for calculations, and when we group/filter we can always display it without the `.0` like this:

```python
print(int(df['year'].min()))  # prints 2004, not 2004.0
```

Try changing `.astype(int)` → `.astype('Int64')` and it should work! 😊

fix this one : print(df[['title', 'year']].head(10))

Thought for 2s

Just change the year column to `'Int64'` before printing so it shows clean whole numbers without `.0`:

```python
df['year'] = df['title'].str.extract(r'(\b(19|20)\d{2}\b)')[0].astype('Int64')
print(df[['title', 'year']].head(10))
```

Suggestions after Identifying Errors

how can california be its own provencee when its so giant

Thought for 1s

Really good observation! You're right — California is **huge** and has dozens of distinct wine sub-regions like Napa Valley, Sonoma, Central Coast, etc.
In this dataset, `province` is the **broadest geographic grouping** used by Wine Enthusiast magazine — so all California wines just get labelled "California" regardless of which sub-region they're from.
The more specific location is in `region_1` — for example:

```
province    → California
region_1    → Napa Valley
region_1    → Sonoma Coast
region_1    → Central Coast
```

For our project we have two options:

Option	Pros	Cons
Use `province` (California)	More wines per group, more reliable averages	Too broad — Napa and Central Valley have very different climates
Use `region_1` (Napa Valley)	More precise, better for climate matching	More missing values (21,247 missing!)

My suggestion: We could use `region_1` for US wines where it's available, and fall back to `province` for other countries.
What do you think — do you want to keep it simple with `province`, or go more detailed with `region_1`? 🍷

Now I have the full picture. This is a sharp methodological observation — let me give a clear answer.

Short answer: No, 2016 is not a good example year — and the reasons you're sensing are real and worth knowing.

Here's exactly what the data shows:

Year	# Reviews	Regions with ≥5 reviews
2012	3,716	16 / 16
2013	3,597	16 / 16
2015	1,713	13 / 16
2016	**522**	**12 / 16**

Decision Log

Agent Contribution	Description	Verification Performed	Decision
Code and baseline pipeline	Generated initial project structure and baseline modelling pipeline.	Reviewed implementation, corrected structural issues, and ensured compatibility with analysis workflow.	**Modified**
Debugging support	Suggested fixes for code errors and implementation issues.	Manually checked outputs and logic before applying fixes.	**Accepted**
Test and documentation drafting	Produced draft documentation and supporting comments.	Edited for clarity, accuracy, and formatting consistency.	**Modified**
EDA and plot suggestions	Proposed exploratory analyses and visualizations.	Performed EDA independently and validated correctness of interpretations.	**Accepted (after validation)**
Dataset usage recommendation	Suggested using all available years of data.	Reviewed data completeness and restricted analysis to relevant years (e.g., 2016 with strongest coverage).	**Rejected / Modified**
Regional feature definition	Treated US states as equivalent to wine regions.	Identified mismatch with realistic viticultural regions and corrected regional grouping.	**Rejected**
Model interpretation methods (PDP, SHAP, GBM)	Suggested interpretability techniques.	Tested relevance against project goals; determined they distracted from predictive focus.	**Rejected**
Climate–season alignment	Initially ignored hemisphere-specific growing seasons.	Researched growing and harvest months by hemisphere and aligned climate data accordingly.	**Modified**
Wine variety handling	Did not account for differences between wine varieties.	Updated dataset and ensured models were run with wine variety included as a feature.	**Modified**